
BISHOP COTTON WOMEN'S CHRISTIAN COLLEGE



INTERNSHIP REPORT – 2026

AREENA MARIYAM WAHEED

Reg No. U18B023S0094

III BSC-NDZ (NUTRITION AND DIETETICS & ZOOLOGY)

Under the supervision of

Ms. Meghana N

Assistant Professor

Department of Home Science



EVALUATION OF INTERNSHIP REPORT

For submission to Dr Manmohan Sing Bengaluru City University

Student	Mentor/Supervisor
Name: Areena Mariyam Waheed	Name: Ms Meghana N
Reg. No: U18BO21S0094	Signature:
Signature:	

EVALUATION SUMMARY

	Project	Viva-voice	Total
Max. Marks	30	20	50
Marks. Allotted			
Name of examiner:	Sign:		



BISHOP COTTON WOMEN'S CHRISTIAN COLLEGE

19,3RD CROSS, C.S.I COMPOUND, MISSION ROAD, BANGALORE-560027

COLLEGE CERTIFICATE

This is to certify that **Ms. Areena Mariyam (U18BO21S0094)** has successfully completed the Internship at “Vikram Aura Hospital” in partial fulfilment of the requirement of Dr Manmohan Singh Bengaluru City University for the award of the degree of Bachelor in Science for the Academic year 2025 2026.

This record of work was carried out by her under the guidance and supervision of college staff.

Signature of HOD:

Signature of Principal

Date:



This is to certify that **Ms. Areena Mariyam Waheed**, studying at Bishop Cotton Women's Christian College, Bangalore. She has successfully completed her 1 month of internship program as "**Intern-Dietitian**" with Vikram Aura Private Limited (Vikram Aura). Her internship tenure was from **9th February, 2026** to **7th March 2026**.

During the span, we found her punctual, handling major responsibilities and very productive.

We wish her all the success in her future endeavours.

Signature of Chief Dietician

Mrs. Affrin Noor

DECLARATION

I, Areena Mariyam hereby declare that this internship report is a record of original work done by me under the guidance of my esteemed guide Mrs. Affrin Noor, Chief dietician, Vikram Aura Hospital, Bangalore. This report is submitted to BISHOP COTTON WOMENS CHRISTIAN COLLEGE in partial fulfilment of the requirements for the internship session during the period of 9thFebruary -7thMarch 2026. It is not formed on the basis for the award of any degree/diploma/ associate ship/fellowship of other similar title to any candidate of any University.

PLACE: BANGALORE

SIGNATURE OF THE CANDIDATE

DATE:

AREENA MARIYAM



BISHOP COTTON WOMEN'S CHRISTIAN COLLEGE

19,3RD CROSS, C.S.I COMPOUND, MISSION ROAD, BANGALORE-560027

DECLARATION

I, **Areena Mariyam Waheed (U18BO21S0094)**, hereby declare that this internship report is my original work carried out under the guidance of Dt Affrin Noor, Senior Dietitian, Vikram Aura Hospital, Bengaluru.

This internship report is submitted to Dr Manmohan Singh Bengaluru City University for the award of the degree of Bachelor in Science. I also declare that this full report is part of it, has not been submitted to any other institution or university for the award of any degree or diploma.

Place: Bengaluru

Name of the Student: Areena Mariyam

Date:

Signature:



BISHOP COTTON WOMEN'S CHRISTIAN COLLEGE

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CERTIFICATE OF THE GUIDE

This is to certify that **Ms. Areena Mariyam (U18BO21S0094)**, student of 3rd BSC (Nutrition and Dietetics, Zoology), has successfully Completed 90 hours of internship in Four weeks including weekends at Vikram Aura Hospital under my guidance, for the partial fulfilment of the requirements for the award of the degree of Bachelor in Science from Dr Manmohan Singh Bengaluru City University.

Place: Bengaluru

Signature of Guide

Date:

ACKNOWLEDGMENT

I would like to express my heartfelt gratitude to the Almighty for blessing me with the strength, patience, and determination to successfully complete my internship. His grace and guidance gave me the confidence to overcome every challenge and continue learning throughout this journey.

I sincerely extend my gratitude to our Principal, Mrs Sunila Kumari, and the Head of the Department, Dr. Marie Kavitha Jayakaran, of Bishop Cotton Women's Christian College for their continuous encouragement, support, and motivation throughout my academic journey. Their guidance has played a significant role in helping me grow both personally and professionally.

I would also like to convey my sincere thanks to Vikram Aura Hospital for giving me the opportunity to undergo my internship in such a reputed institution. The experience and exposure I gained during my training greatly enhanced my knowledge in Clinical Nutrition and Dietetics and strengthened my passion for this profession.

I am deeply thankful to our respected Chief Dietician, Mrs. Affrin Noor, for her valuable guidance, constant support, and inspiring mentorship throughout the internship period. Her expertise, encouragement, and constructive feedback helped me improve my skills and gain confidence in my professional abilities.

"I would like to express my heartfelt gratitude to Ms. Meghana. N for her constant guidance, patience, and encouragement throughout the preparation of my internship report. She devoted her valuable time in carefully reviewing my work, correcting mistakes, clarifying my doubts, and guiding me in improving the content and overall presentation of the report. Her constructive suggestions and attention to detail helped me identify errors, make necessary corrections, and complete the report in a more organized and professional manner. Her continuous support and motivation gave me the confidence to perform my work sincerely and efficiently. I am truly grateful for her mentorship, kindness, and unwavering support, which played an important role in the successful completion of my internship report.

Most importantly, I would like to thank my beloved parents and family for their endless love, sacrifices, prayers, and unwavering support. Their belief in me has always been my greatest strength and motivation to strive for success.

Finally, I extend my gratitude to everyone who supported and encouraged me during this journey. Each experience, lesson, and piece of guidance has contributed to my growth and learning, and I will always remain grateful for the opportunity to learn under such wonderful mentors and institutions.

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HISTORY OF FORTIS HOSPITAL – MERGED TO VIKRAM AURA HOSPITAL



Vikram Hospital, located in the arterial Richmond Road, Bengaluru is now being unveiled in its new avatar as a multi-speciality hospital. The hospital was inaugurated by Nalpad Ahmed Haris, honourable member of the Karnataka legislative assembly in the presence of Akshay Oleti, Business head, Fortis hospitals and Raghuram M.D, facility director, Vikram hospital, Richmond Road with a strong team of 50 expert clinicians across specialities, the facility will cater to the residents and people in and around Richmond Town. The 85-bedded medical facility, which has always been on the forefront catering exclusively to women's health, has now stepped-up its services and extending it to other medical specialities including Internal Medicine, General Surgery, Obstetrics and Infertility, Paediatrics – ICU & NICU, Orthopaedics, Urology, Oncology, Cardiology and many more.

Fortis Health Care has always been synonymous with providing excellent medical services, well-equipped labs, diagnostic centres, and advanced surgical theatres – all backed by exceptionally trained doctors, and surgeons of global standards. Our hands-on paramedical staff and nursing work in tandem with various departments, and doctors for ensuring seamless comfort, and care all in the pursuit of alleviating pain to the patients and bringing smiles to their families.

Besides Emergency Services that run round-the-clock, the hospital also has Outpatient Facility, Blood Storage Bank, Radiology, Medical ICU, Tertiary Level 3 NICU and PICU. The affordable packages offered under Preventive 7 Health Check-up help you gauge your health in detail and make necessary lifestyle changes for healthy living.

The stay options are devised to suit the requirements of patients from various walks of life. The room options available are General Ward, Twin Sharing, Single Deluxe, Suite and Presidential Suite.

This is the only hospital that contains “Amaara Human Milk Bank” which is a non-profit Human Breast Milk Bank in the state of Karnataka.

The logo of the Fortis brand projects is a symbol of human values such as trust, ethics, service, and quality. The union of the green colour hands and a red dot represent Fortis path to healthcare. The green colour of hands represents compassion, nurturing, generosity, wellbeing, and health. The red dot stands for energy, spirituality, courage, and a symbol of good luck.



VALUES OF VIKRAM AURA HOSPITAL

Vikram Aura Hospital Richmond Road, located in Bangalore, India, is known for its commitment to providing high-quality healthcare services. The hospital upholds several core values that guide its operations and patient care.

VISION

To become India's most trusted and innovative healthcare network, setting the global benchmark for clinical excellence, technology integration, and patient-centric experiences.

MISSION

Enhance the healthcare journey with a technology forward, compassionate, and premium experience tailored to families and individuals

CORE VALUES

P – Precision

I – Innovation

C – Comfort

E – Exclusivity

Scope of services

Speciality offered

- Internal medicine
- Orthopaedics
- General /Minimal Access surgery
- Urology
- Neurosciences
- Medical oncology
- Surgical oncology
- Non- invasive & invasive cardiology
- Peripheral vascular surgery
- Diabetology & Endocrinology
- ENT
- Dermatology
- Anaesthesia & pain management

Other services

- Emergency (24×7)
 - Cardiac intensive care unit
 - Intensive care unit
 - Nutrition and dietetics
 - Physiotherapy
 - Pharmacy (24×7)
-

Diagnostic facilities

- Ultrasonography
- Portable X-Ray/Fixed X-Ray
- CT scan
- Mammograph
- ECG
- Echocardiogram
- Treadmill Test (TMT)
- Pulmonary Function Test

Laboratory Services

- Clinical Biochemistry
 - Clinical Pathology
 - Haematology
 - Clinical Microbiology and Serology
 - Histopathology
 - Cytology
 - Molecular Biology
-

DIETARY DEPARTMENT

Diet plays a very important role in the treatment of the disease. Keeping this in mind the management decided to start fully pledged department of dietetic in the year of 2015 to bring world class health care to the people.

It plays a crucial role in providing nutritious and well-balanced meals to patients, catering to their specific dietary needs based on their medical conditions and treatment plans.

Patients with specific medical conditions or dietary restrictions may require specialized diets. The dietary department is equipped to provide tailored meal options for individuals with diabetes, heart disease, food allergies, and other health concerns.

The department is staffed with qualified dietitian and culinary professionals who work together to ensure that patients receive optimal nutrition during their stay at the hospital.

Vikram Aura Hospital Richmond Road's dietary department adheres to stringent quality standards to uphold the nutritional well-being of its patients. This includes maintaining hygiene and food safety protocols in meal preparation, utilizing high-quality ingredients, and monitoring the nutritional content of meals served.

Presently the dietary is led by the chief dietician Mrs. Affrin Noor.

FUNCTIONS OF DIETARY DEPARTMENT

On the beginning of the day the dieticians receive daily diet lists. It includes name, bed number, ward number, diagnostic condition and diet (Normal, diabetic, soft, renal etc...) of the patient. The diet is prescribed by the doctor and it was sent by night duty nurses from the respective ward to the Dietary.

The mid-morning and mid-evening beverage summary according to the condition of the patient is made and the dietician leaves for the round. During the rounds, changes of the diet need and likes of the patient are noted.

After completing the rounds, dieticians come from the ward and makes changes on the diet slip for each patient according to the present dietetics' status. Once all the diet slips are prepared, diet summary is made. It's given to the kitchen for preparation. In the diet slip bed number, diet and order will be mentioned.

According to this slip, lunch setting is done in the presence of dieticians to make sure that correct diet is going to respective patients. The diet is set in the trolley in ward wise order and given to the patient with the help of the diet slip.

Tea or coffee service is done according to the list given to the stewards by the dietician. A dinner summary is made with the diet slip and given to the kitchen for preparation. Setting of dinner will be done and served to the patient. Bed time milk is given to most the patients.

During duty time phone calls, slip regarding diet changes, new admissions, transfer, NPO over are also received and taken necessary care.

ROLE OF DIETETICS AND DIETICIANS IN THE HOSPITAL

Dietetics and dietitians play a crucial role in the overall management of patients in a hospital. Their expertise is essential in ensuring that patients receive the proper nutrition to support their recovery and maintain their health.

Dietitians are responsible for assessing and evaluating the nutritional needs of each patient.

Dietitians provide nutrition counselling to patients and their families, educating them on the importance of a balanced diet and the role it plays in promoting health and preventing diseases.

Dietitians work closely with the foodservice department to develop menus that meet the nutritional needs of patients and adhere to hospital guidelines.

Dietitians monitor the quality and safety of the food served to patients.

Dietitians are responsible for monitoring the progress of each patient and making necessary adjustments to their meal plans.

Dietitians work closely with other healthcare professionals, such as doctors, nurses, and pharmacists, to ensure that the patients' nutritional needs are integrated into their overall treatment plan.

Dietitians help to promote healthy life style and control various diseases.

ROLES AND RESPONSIBILITIES OF STAFFS

CHIEF DIETICIAN

- Plans, develops, controls and evaluates food service system.
- Develops short- and long-range department programs.
- Consistence with departmental and organizational policies.
- Manages and controls financial resource and recommended budget programs.
- Establishes and maintains standards of food production and services, sanitation, safety and security.
- Maintains effective return and verbal communication and public relations inter and intra departmentally.
- Plans, conducts and evaluates orientation and in service educational programs.
- Maintain records on quality, quantity, post and frequency of food given to the patient.
- Goes on hospital rounds and interact with the patients and asks them if they need any changes in the menu.
- Check the case file of the patient to see if the doctors prescribed any changes in the diet.
- To assess the nutritional status of patients in health and disease.
- To communicate with the doctors and monitor implantation of nutrition care plan.
- To document all aspects of nutritional care.
- To communicate effectively and share knowledge with other member of health team.
- Guide the internship students and supervising the workers.

COOKS

- Prepare the menu according to the weekly and daily menu chart.
- Gets the ingredients beforehand from the store and the preparation is done.
- He should be trained enough to make the different therapeutic diet like salt free, renal, bland diet etc.
- He should use clean hands and hygienic method of cooking.
- He must cook on time and quickly, as there should be no delay for the patient to receive food.

TROLLEY BOYS

- Firstly, they should arrange the set menus, according to floor and room wise.
 - He should cleanly and neatly arrange the tray, room wise in the trolleys.
 - The plates and glasses should be arranged so that it's served hot.
 - He must move smoothly with the trolley to avoid collision and scattering.
 - He must serve the right tray to the right room, serving the tray of other rooms would lead to wrong diet to wrong food intake and complication.
-

MAIDS

- She should sweep and mop the kitchen and other areas periodically.
- She should particularly use disinfectants in the water to mop.
- She must wash the dishes and the soiled utensils using washing powders for periodic use by the cooks. She must clean the work tables and other equipment's.
- To collect all the dirty plates and deliver at the washing area.
- The patient's plates and canteen plates are collected separately.

DUTIES OF INTERNS

- Internship is the periods where the students are given opportunity to put the theoretical knowledge into practice.
- The main purpose of internship in dietetics for a period of one month is to provide an opportunity for the students to work under a registered/chief dietician in a well-established hospital to gain practical knowledge and experience and to understand dieticians in a better way.
- Vikram Aura Hospital also provides internship facilities under the guidance of chief dietician Mrs. Affrin Noor.

HOSPITAL ROUNDS

- Interns go for rounds along with assigned assistant dieticians. From the nursing station they collect changes in the diet.

LUNCH AND DINNER SETTING

- The interns then prepare the lunch and dinner slips accordingly to the timing for each and passes it the trolley boys who arrange the food as suggested. The interns also check the food for its quality, texture, taste and hygiene.
-

DIET COUNSELING

Diet counselling, also known as nutrition counselling, is a process in which a qualified healthcare professional, such as a chief/registered dietitian or nutritionist, provides guidance and support to individuals regarding their dietary habits and nutrition-related goals. The aim of diet counselling is to help individuals make informed choices about their food intake and develop healthier eating patterns to achieve and maintain optimal health. It empowers individuals to take charge of their own health by making informed choices about their diet. It provides the knowledge, skills, and support necessary for individuals to adopt healthier eating habits, improve their overall well-being, and prevent or manage various health conditions.

Key Aspects of Diet Counselling

- **Assessment:** The diet counsellor begins by conducting an assessment of the individual's current dietary habits, health status, medical history, lifestyle, and specific nutritional needs.
- **Goal Setting:** Based on the assessment, the counsellor works with the individual to establish realistic and personalized goals. These goals may include weight management, managing chronic conditions (e.g., diabetes, hypertension), improving athletic performance, addressing nutrient deficiencies, or promoting overall wellness.
- **Education and Guidance:** The counsellor provides evidence-based information and education about nutrition, including the role of various nutrients, portion control, food groups, and the impact of dietary choices on health.
- **Individualized Meal Planning:** The counsellor assists individuals in developing personalized meal plans that meet their specific needs and goals.
- **Behavioural and Lifestyle Modification:** Diet counselling often focuses on modifying eating behaviours and promoting healthy lifestyle habits. The counsellor helps individuals identify and address barriers to healthy eating, such as emotional eating, food cravings, or lack of time for meal preparation.

IN-PATIENT COUNSELING

- In patient counselling is the counselling given to the patients who are admitted in the hospital.
- The dietitians perform screening and prescribes diet for the patient
- The dietitians give counselling for those patients for whom the doctor has advised diet counselling according to the patient's diagnosis.
- Diet charts are provided by the dietitian, which contain the list of food items which are to be consumed or avoided or restricted by the patient.

OUT PATIENT COUNSELING

- Out-patient counselling is given to the patients who come for consultation only.
 - The patients are referred to the dietitians by the physician for counselling.
 - The dietitian counsels the patients according to the disease condition of the patient.
 - Diet charts are provided by the dietitian which contains the list of food items which are to be consumed or avoided by the patient.
 - The dietitian counsels the patients with aim to give guidance to maintain a healthy body by eating the right nutritious food.
-

KITCHEN

The kitchen at Vikram Hospital is designed to cater to the dietary needs of patients, ensuring that they receive nutritious and delicious meals throughout their stay. This kitchen is staffed by skilled professionals who follow strict hygiene and safety protocols to maintain the highest standards of food preparation.

One of the key features of the kitchen is its capacity to cater to the diverse dietary requirements of patients. The hospital recognizes the importance of a well-balanced diet in a patient's recovery process and works closely with doctors and nutritionists to create personalized meal plans. These meal plans take into consideration the patient's medical condition, allergies, and preferences, ensuring that they receive the appropriate nutrition for their specific needs.

In addition to patient meals, the kitchen also caters to the needs of visitors and hospital staff.

The kitchen is regularly inspected and maintained by professional cleaning staff to ensure that it remains a safe and sanitary environment.

PANTRY

The pantry at Vikram Aura Hospital Richmond Road is an essential facility that provides food and beverages to patients, visitors, and staff within the hospital premises.

It plays a crucial role in ensuring that individuals have access to nourishing meals and refreshments during their time at the hospital.

The pantry also offers freshly brewed coffee and tea, which are popular choices among visitors and staff.

The hospital recognizes the importance of a well-stocked pantry in providing a pleasant experience to its patients, visitors, and staff, and has taken necessary measures to ensure that it meets their expectations.

Its role in maintaining quality, adhering to hygiene standards, and contributing to patient care underscores its significance within the hospital environment.

EQUIPMENTS IN KITCHEN AND PANTRY

MAJOR	MINOR
Bain Marie	Sambar and Rassam bowl
Fridge	Plate small
Mixer	Big Plate
Grinder	Kadai
Oven	Bucket
Griller	Cooker
Milk Boiler	Sauce Pan
Trolley	Steel Pan
Table	Steel Drum
Washbasin	Coffee can
Store	Chopping Board

FOOD SERVICE AND DELIVERY

DIETARY TIMINGS

SESSION	PREPERATION	TASTING
Early morning	6:10 am	-
Breakfast	7:30 am	8:00am
Mid-morning	10:00am	10:10pm
Lunch	12:00pm	12:15 pm
Snacks	3:00pm	-
Soup	5:30pm	5:45
Dinner	6:30 pm	7:00pm
Bed Time	8:45 pm	8:55pm

METHODS USED FOR COOKING

- Boiling
 - Steaming
 - Shallow frying
 - Deep frying
 - Roasting
 - Pressure cooking
 - Gravy
 - Leaching
-

RECORDS USED

1.DIET SHEET -

Consists the details of admitted patients along with their room number, UHID number, type of diet and their specific food allergies. Dietician takes these sheets while going on rounds and attaches it with the diet card after preparing the cards.

2.FOOD TRAY CHECKLIST -

In this book, the patients room number along with their names and the food sent to the patient is entered

3. COMPLAINT REGISTER -

In this register the patient's complaints are recorded. This helps to address customer dissatisfaction, improve their services or products, and maintain a positive reputation.

4.TASTING RECORD -

Tasting book is used to record the quality of food such as taste, texture, aroma and consistency of the menu every day. Its recorded by the dieticians after tasting the food.

FOOD DISPATCH SYSTEM

The foods are set in hot boxes and dispatched through service trolleys.

The fluids are despatched either in the service trolley or they are dispatched off in tray by the service person.

SANITATION AND HYGIENE IN KITCHEN AND DINING

The kitchen of Vikram Aura Hospital Richmond Road adheres to strict standards of cleanliness and hygiene to ensure the health and well-being of patients, staff, and visitors.

Sanitation and hygiene practices are as follows:

- Use of hair caps and gloves by the employees.
- Regular cleaning and sanitizing of all surfaces, equipment, and utensils.
- Storing raw and cooked food separately to prevent cross-contamination.
- Properly disposing of waste and ensuring no food waste is left out for pests.
- Implementing a pest control program to prevent infestations.
- Training and supervision of kitchen staff to ensure they follow proper hygiene practices. ▪ Ensuring proper ventilation and temperature control in the kitchen.
- Use of disposable cups.

PURCHASE AND STORAGE

FOOD STORAGE

Vikram Aura Hospital follows best practices for food storage, including temperature control, segregation of raw and cooked foods, labelling and dating of items, and regular monitoring of inventory to prevent spoilage or waste. The hospital's kitchen facilities are equipped with refrigeration units, freezers, and storage areas designed to meet food safety standards.

The perishable food products such as fruits and vegetables are purchased on a daily basis depending upon the stock in hand.

The non-perishable food products such as cereals and grains are bought on a weekly basis.

PURCHASE

The hospital's food services department is responsible for procuring, storing, and preparing food.

The food purchase process involves several steps to ensure the procurement of high-quality and safe food products. The hospital likely has established relationships with approved vendors and suppliers who adhere to stringent quality control measures. When purchasing food items, the hospital's procurement team consider factors such as supplier reputation, product quality, pricing, and adherence to food safety regulations.

A list of required material is prepared according to shortage, then the list is sent to the purchase department. The list of the items is sent back to the store manager, then the purchasing is done.

The schedule of purchasing is on the 30th of every month.

The dietary adopts three kinds of purchasing:

-
- (a) Daily purchases
 - (b) Bulk Purchases and
 - (c) Emergency purchases

There were different storage conditions available according to the type of food.

DRY STORAGE: All non-perishable food items such as cereal, pulses, spices, sugar, etc. are stored separately in different cans.

VEGETABLE STORAGE: Vegetables are kept in store rooms in separate shelves. The room is impervious, clean, and free of excessive light and termites.

COLD STORAGE: Perishable food items such as non-vegetarian items, milk, milk products, and cream etc. are kept in cold storage. Refrigerator, freezers and deep cold freezers are used.

First in and first out procedure is followed.

TYPES OF THERAPEUTIC DIETS

The Vikram Aura hospital's dietary department provides the following special therapeutic diets to the patients:

- Regular normal diets
 - Bland diets
 - Low fat diets
 - Soft diet
 - Clear liquid diet
 - Liquid diet
 - Diabetic diet
 - Renal diet
 - Salt restricted diet
 - RT feed
-

AIM

To gain practical knowledge and hands-on experience in the field of Clinical Nutrition and Dietetics.

To apply theoretical knowledge learned during academics in a real hospital setting.

To understand the role and responsibilities of a clinical dietitian in patient care and disease management.

To observe the dietary management of various diseases such as diabetes, hypertension, chronic kidney disease, liver disorders, gastrointestinal disorders, and cardiovascular diseases.

To improve communication and interpersonal skills while interacting with patients and healthcare professionals.

To understand the importance of teamwork, professional ethics, and patient-centred care in a hospital environment.

To gain exposure to hospital food service management and clinical nutrition practices.

To enhance confidence, professional competency, and decision-making skills in the field of nutrition and dietetics.

To strengthen my passion and interest in Clinical Nutrition and Dietetics as a future profession.

To gain overall clinical exposure and experience required for future professional practice.

OBJECTIVES -

To observe and understand the functioning of different departments in the hospital.

To learn the process of nutritional assessment and diet planning for patients.

To understand therapeutic diets and their importance in disease management.

To gain knowledge about the nutritional requirements of patients with various medical conditions.

To develop skills in patient counselling and nutrition education.

To learn about enteral and parenteral nutrition support used in clinical settings.

To understand food service management and hygiene practices followed in hospitals.

To improve professional communication skills while interacting with patients and healthcare professionals.

To learn the importance of accurate documentation and maintenance of diet records.

To observe the role of nutrition in recovery, health promotion, and disease prevention.

WORK EXPERIENCE

Work Experience – Week 1 & Week 2

During the first and second week of my internship at Vikram Aura Hospital under the guidance of Dr. Affrin Noor, I was introduced to the hospital environment, nutrition department activities, and the responsibilities of a clinical dietitian. The internship helped me gain practical exposure to hospital nutrition care and patient management.

On the first day, I attended the hospital induction program where I understood the basic terms and procedures used in the hospital dietetics department. I was also taken for hospital rounds to observe the functioning of different wards and departments. This helped me become familiar with the clinical setup and the role of nutrition services in patient care.

On the second day, I learned about nutritional assessment forms and formats in detail. I observed how dietitians prepare patient reports and provide nutrition counselling to patients based on their medical condition and dietary needs.

On the third day, I participated in the preparation of nutritional assessment sheets for patients. I also observed patient assessment procedures, diet card preparation, and pantry visits. Food tasting and tray checking activities were conducted to ensure food quality and hygiene.

On the fourth day, I observed diet counselling and diet planning for patients with different medical conditions. I also learned about food checking procedures, food testing methods, and ward food distribution systems.

On the fifth day, I attended ward and ICU rounds and observed different feeding practices followed for patients. I also interacted with the medical representative of Ensure and understood the importance of nutritional supplements in clinical nutrition management.

On the sixth day, I observed clinical assessment procedures in the intensive care unit. I studied biochemical parameters and their importance in therapeutic diet planning and nutritional intervention.

On the seventh day, I was introduced to case study preparation and was assigned a case study for further work. I also received guidance regarding dietary interventions for patients.

During the second week, I continued participating in ward rounds, ICU rounds, nutritional assessment, food tray checking, and case study documentation. I collected details for the assigned case study and observed how nutrition care plans are prepared for hospitalized patients. I learned about food tray checking procedures, patient counselling, and documentation practices used in the hospital.

I also assisted the chief dietitian in ward activities and documentation work. I gained knowledge regarding nutrition data entry, diet planning, and nutritional assessment techniques. Under supervision, I observed counselling sessions conducted for patients requiring high-protein diets and therapeutic nutrition support.

Overall, the first two weeks of my internship provided me with valuable exposure to clinical dietetics, patient interaction, therapeutic diet planning, nutritional assessment, and hospital food service management. It improved my understanding of professional dietetic practices and enhanced my communication and observational skills.

Work Experience – Week 3 & Week 4

During the third and fourth week of my internship at Vikram Aura Hospital, I gained more practical experience in clinical nutrition assessment, diet planning, case study preparation, and ICU nutritional management. These weeks helped me improve my practical knowledge and confidence in handling nutrition-related activities in a hospital setting.

At the beginning of the third week, I continued nutritional assessment procedures and observed diet counselling sessions under the supervision of the clinical dietitian. I actively participated in food tray checking to ensure that patients received the prescribed therapeutic diets correctly.

I also worked on my assigned case study by collecting patient details, medical history, dietary intake, and biochemical data. This activity helped me understand the importance of accurate documentation and individualized nutrition care planning.

During this period, I observed diet planning procedures for patients with different disease conditions. I learned how dietitians calculate calorie and protein requirements according to the patient's condition, age, weight, and medical diagnosis.

I was introduced to RTF (Ready to Feed) calculations and learned the method of calculating calorie intake and nutritional requirements for critically ill patients. I also learned about water intake calculations and individual calorie count preparation for patients admitted in the ICU and wards.

Ward rounds and ICU rounds were conducted regularly, where I observed the nutritional management of patients receiving enteral and therapeutic feeding. I also learned about the importance of monitoring food intake, nutritional status, and patient response to dietary modifications.

In the fourth week, I continued assisting in nutritional assessment, food tray checking, and documentation work. I observed diet counselling sessions for patients requiring therapeutic and modified diets. I also participated in nutrition data entry and diet planning assessment procedures.

Case study preparation continued throughout the week, helping me understand the practical application of nutrition assessment methods and therapeutic diet planning. I observed how dietitians communicate with patients and provide dietary guidance according to their health conditions and treatment requirements.

The internship experience during the third and fourth week enhanced my understanding of clinical nutrition care, ICU nutrition management, nutritional calculations, and patient counselling. It also improved my practical skills in diet assessment, documentation, communication, and teamwork within a hospital environment.

Overall, the internship at Vikram Aura Hospital was a valuable learning experience that helped me bridge the gap between theoretical knowledge and practical application in the field of Nutrition and Dietetics.

LEARNING OUTCOMES

During my internship at Vikram Aura Hospital, I gained valuable practical exposure in the field of Clinical Nutrition and Dietetics. This internship helped me understand how the theoretical concepts learned during academics are applied in a real hospital setting. It enhanced my knowledge, practical skills, and professional confidence.

Through this internship, I learned how to perform nutritional assessments using anthropometric measurements, dietary history, biochemical parameters, and clinical information. I gained knowledge about planning and modifying therapeutic diets according to the nutritional needs and medical conditions of patients. I also understood the dietary management of diseases such as diabetes mellitus, hypertension, chronic kidney disease, liver disorders, gastrointestinal diseases, and cardiovascular disorders.

The internship also improved my communication and counselling skills. By observing interactions between dietitians and patients, I learned the importance of educating patients about healthy food habits and therapeutic diets in a simple and effective manner. It also helped me develop confidence while interacting with patients and healthcare professionals.

In addition, I gained exposure to hospital food service management, hygiene and sanitation practices, maintenance of diet records, and the importance of teamwork in patient care. I understood the professional responsibilities of a clinical dietitian and the importance of discipline, punctuality, ethics, and empathy in a healthcare setting.

Overall, this internship was a great learning experience that improved my practical knowledge, observational skills, and professional competence. It strengthened my interest in Clinical Nutrition and Dietetics and prepared me for future professional practice.

Another important learning outcome of my internship at Vikram Aura Hospital was the development of professional attitude and clinical observation skills. During the internship, I learned how healthcare professionals work together to provide quality patient care and how nutrition plays an essential role in improving patient outcomes. I gained exposure to case discussions, patient monitoring, diet planning, and hospital routines, which helped me understand the practical responsibilities of a dietitian. The internship also taught me the importance of time management, accuracy in documentation, maintaining patient confidentiality, and following professional ethics in a clinical setting. By observing different patient cases and therapeutic approaches, I was able to improve my analytical thinking and decision-making abilities. This experience not only enhanced my academic knowledge but also helped me become more responsible, confident, and prepared for my future career in Clinical Nutrition and Dietetics.

CONCLUSION

The internship at Vikram Aura Hospital was a highly informative and enriching experience that provided me with practical exposure in the field of Clinical Nutrition and Dietetics. It helped me apply the theoretical knowledge gained during my academic studies in a clinical environment and understand the important role of nutrition in patient care and disease management.

Throughout the internship, I gained knowledge in nutritional assessment, therapeutic diet planning, patient counselling, and dietary management of various diseases. I was able to observe the role of dietitians in promoting patient recovery and improving overall health through proper nutrition care. The internship also enhanced my communication skills, confidence, and professional approach while interacting with patients and healthcare professionals.

responsibilities in a clinical setting.

The guidance and support provided by the Chief Dietician, doctors, and other staff members were instrumental in enhancing my learning experience. Their encouragement and mentorship helped me build confidence, improve my skills, and develop a professional approach towards my work.

This internship also helped me improve my communication skills, enabling me to interact effectively with patients and healthcare professionals. I learned the importance of empathy, patience, and clarity while dealing with patients, which are essential qualities in the healthcare field.

Moreover, the experience taught me the value of discipline, responsibility, and teamwork in a professional environment. It encouraged me to stay committed, adaptable, and open to continuous learning.

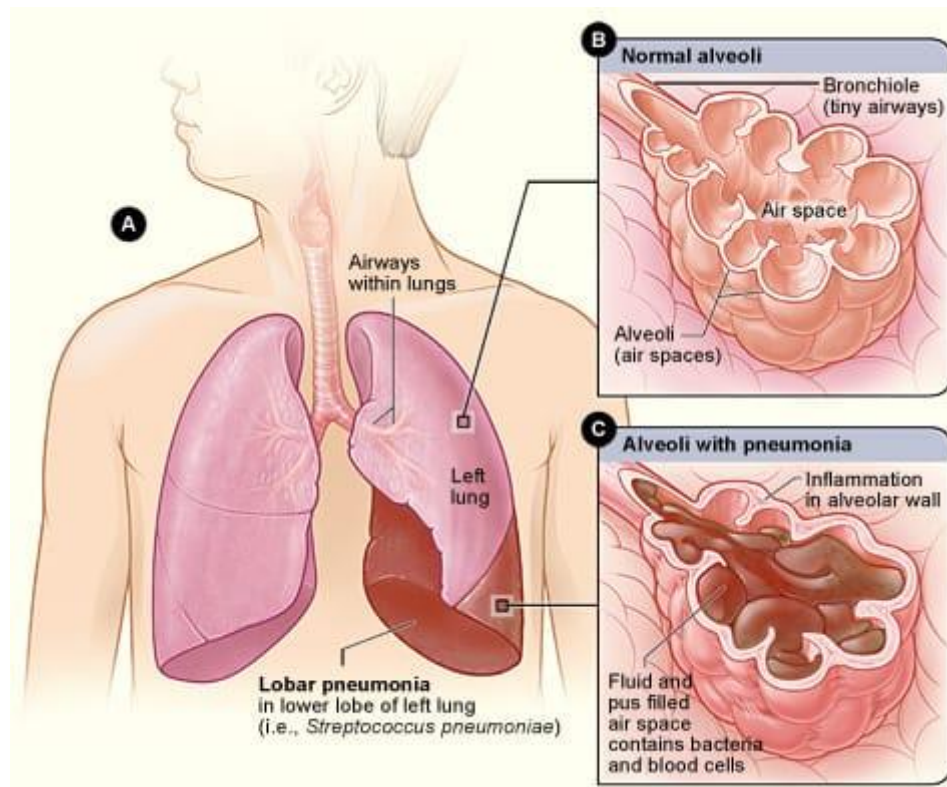
In conclusion, this internship has been a meaningful and transformative experience that has contributed significantly to my personal and professional growth. It has strengthened my confidence and dedication. I am truly grateful for the opportunity and the knowledge I have gained during this period.

This internship strengthened my interest and passion for Clinical Nutrition and Dietetics and motivated me to continue learning and improving my knowledge in this field. Overall, the internship was a meaningful and rewarding experience that contributed to both my academic and professional growth and prepared me for future opportunities in healthcare and nutrition practice.

CASE STUDIES

CASE STUDY -1

PNEUMONIA



INTRODUCTION:

Pneumonia is a common illness and remains a major cause of morbidity and mortality worldwide. According to estimates cited in the article published in *The Lancet*, about 450 million cases of pneumonia occur annually, resulting in approximately 4 million deaths each year. The burden is greatest in children younger than 5 years in developing countries and in elderly individuals in developed countries.

Many microorganisms can cause pneumonia, including both bacteria and viruses. In recent years, increasing attention has focused on the role of respiratory viruses as important pathogens. Widespread use of *Haemophilus influenzae* type b and pneumococcal conjugate vaccines has changed the epidemiology of childhood pneumonia and raised the possibility that viruses now account for a larger proportion of cases. The emergence of severe viral infections such as severe acute respiratory syndrome (SARS), avian influenza A (H5N1), and the 2009 pandemic influenza A (H1N1) has further emphasized the importance of viral causes of severe pneumonia. In patients with underlying conditions such as diabetes mellitus, pneumonia may be more severe due to altered host defence mechanisms and increased susceptibility to infection

ETIOLOGY:

Viral pneumonia is caused by a wide range of respiratory viruses, and modern PCR-based diagnostics have significantly increased detection rates compared to conventional methods.

1. Major Viral Causes in Children

In children, viruses are identified in approximately 49–67% of community-acquired pneumonia cases when PCR methods are used.

The most frequently identified viruses are:

- Respiratory syncytial virus (RSV) – the leading viral cause, especially in infants and severe cases
- Human metapneumovirus
- Human bocavirus
- Adenovirus
- Coronaviruses (229E, OC43, NL63, HKU1)
- RSV remains the predominant viral cause of severe pneumonia globally in young children.

2. Major Viral Causes in Adults

- In adults, viruses are detected in about 15–56% of cases when PCR is used, with approximately one-third of community-acquired pneumonia cases associated with viral infection.
- The most important viruses include:
- Influenza viruses (most common)
- Rhinoviruses
- Respiratory syncytial virus
- Human metapneumovirus
- Coronaviruses

3. Emerging and Severe Viral Causes

Important epidemic and pandemic viruses:

- SARS Corona virus
 - Avian influenza A (H5N1)
-

-
- Pandemic influenza A (H1N1) 2009

These viruses are associated with severe viral pneumonitis and high mortality.

4. Viral–Viral Coinfection

- Dual or multiple viral infections are common, particularly in children:
- 10–20% of pneumonia cases involve more than one virus.
- Human bocavirus is frequently detected alongside other viruses.
- The clinical significance of multiple viral detection is still uncertain.

5. Viral–Bacterial Coinfection

- Mixed viral-bacterial infections are common:
- Up to 45% of childhood pneumonia cases show evidence of both viral and bacterial infection.
- In adults, mixed infections occur in approximately 14–15% of cases.
- The most common bacterial partner is *Streptococcus pneumoniae*.
- These coinfections may lead to more severe disease and higher mortality

PATHOPHYSIOLOGY:

Pneumonia begins with infection of the respiratory epithelium by respiratory viruses. These viruses infect epithelial cells of the bronchi, bronchioles, and alveoli. In more severe cases, infection extends deeply into the lung parenchyma and involves alveolar epithelial cells.

1. Viral Infection and Epithelial Injury

- Respiratory viruses replicate within epithelial cells, causing cellular injury and destruction.
 - For example: Respiratory syncytial virus infects both bronchial and alveolar epithelium.
 - Rhinovirus antigen has been demonstrated in alveolar epithelial cells and macrophages.
 - Severe infections such as SARS, avian influenza (H5N1), and pandemic H1N1 are associated with extensive pneumocyte damage
-

➤ 2. Inflammatory Response

The infection triggers an inflammatory response characterized by:

- Lymphocytic infiltration
- Accumulation of monocytes and macrophages
- Interstitial inflammation
- The most typical histopathological finding in viral pneumonia is interstitial pneumonitis, where lymphocytes infiltrate the alveolar septa. This inflammation leads to thickening of the alveolar walls and impaired oxygen exchange.

3. Alveolar Damage and Severe Disease

- In severe viral infections:
- Diffuse alveolar damage
- Desquamation of pneumocytes
- Hyaline membrane formation
- Alveolar haemorrhage
- Necrotizing bronchiolitis
- These pathological changes impair gas exchange and may progress to acute respiratory distress syndrome (ARDS)

4. Viral–Bacterial Interaction

Viral damage to respiratory epithelium facilitates bacterial invasion, especially by

- *Streptococcus pneumoniae* and *Staphylococcus aureus*. Mixed viral–bacterial infections are associated with more severe inflammation and worse outcomes.
 - Pathophysiology in a Patient with Diabetes
 - Impaired immune response can reduce the ability to control viral replication.
 - Epithelial damage combined with weakened host defence increases susceptibility to secondary bacterial infection.
-

SYMPTOMS:

General Symptoms

- Fever (often lower than in typical bacterial pneumonia; sometimes $<38.5^{\circ}\text{C}$ in children)
- Gradual onset of illness
- Malaise and fatigue

Respiratory Symptoms

- Cough
- Wheezing (especially in young children)
- Rhinitis (runny nose)
- Chest recession (in children)
- Tachypnoea (increased respiratory rate)

Viral Cause (especially in children)

- Wheezing
- Lower-grade fever compared to bacterial pneumonia
- Intestinal infiltrates on chest radiograph

Severe Cases May Present With:

- Respiratory distress
 - Hypoxia
 - Bilateral interstitial or alveolar infiltrates on imaging
 - In pandemic influenza or SARS: rapid progression to severe respiratory failure
-

DIAGNOSIS:

Diagnosing the cause of pneumonia remains challenging despite advances in diagnostic methods. No single clinical algorithm can clearly distinguish viral from bacterial pneumonia.

1. Clinical Assessment

- Several variables may help suggest viral or bacterial cause, but none are definitive:
- Age (viral more common in young children)
- Speed of onset of illness
- Presence of viral epidemics in the community
- Symptoms such as wheezing (more suggestive of viral cause)
- However, clinical signs and symptoms overlap significantly.

2. Radiological Diagnosis

- The American Thoracic Society recommends that diagnosis of pneumonia should be made on the basis of chest radiography.
- **Interstitial infiltrates** are generally believed to suggest viral pneumonia.
- **Alveolar (lobar) infiltrates** are more commonly associated with bacterial pneumonia
both viruses and bacteria can produce either pattern, so radiographic findings are not definitive.

3. Laboratory Diagnosis

A. Detection of Viruses

- Laboratory diagnosis of viral pneumonia relies on detection of virus or viral antigen in:
 - **Upper respiratory specimens** (nasopharyngeal aspirates or swabs)
 - **Lower respiratory specimens** (induced sputum, bronchoalveolar lavage)
 - Diagnostic methods include:
 - Virus culture
 - Immunofluorescence microscopy
 - Serological testing (paired serum samples)
 - PCR (polymerase chain reaction)
 - PCR-based methods are 2–5 times more sensitive than conventional techniques and have significantly improved detection of respiratory viruses
-

4. Biomarkers

- Certain laboratory markers may help differentiate viral from bacterial pneumonia:
- White blood cell count
- C-reactive protein (CRP)
- Procalcitonin
- Generally:
- Lower CRP and procalcitonin levels suggest viral infection
- Higher levels support bacterial infection.
- However, none of these markers have sufficient sensitivity or specificity to be used alone.

5. Challenges in Diagnosis

- Lower respiratory samples are difficult to obtain.
 - Viral detection in the nasopharynx may represent colonization rather than true pneumonia.
 - Many studies focus on hospitalized patients and may not represent mild disease.
 - Causative agents remain unidentified in many patients.
-

TREATMENT:

1. Antibiotic Treatment

- Some experts recommend antibiotics for all patients, because bacterial infection cannot be reliably excluded.
- The British Thoracic Society suggests that antibiotics may be withheld in young children with mild illness when a viral cause is likely.
- A randomized placebo-controlled study in children showed no significant benefit of antibiotics in mild pneumonia during an RSV epidemic.

2. Antiviral Treatment

A. Influenza A and B

- **Neuraminidase inhibitors** (e.g., oseltamivir, zanamivir) are effective when started early.
 - They reduce duration of symptoms and may reduce complications such as pneumonia.
 - They are recommended for hospitalized influenza patients, even if symptoms began more than 48 hours earlier
 - Pandemic H1N1 virus remains susceptible to neuraminidase inhibitors.
 - **Cidofovir** has been used for severe adenovirus pneumonia.
 - **Acyclovir**
 - is used for varicella pneumonia.
 - **3. Corticosteroids**
 - Use of corticosteroids is controversial:
 - Not effective in respiratory syncytial virus infections.
 - Not recommended in influenza due to possible prolonged viral shedding and increased mortality.
 - Some benefit has been suggested in varicella-zoster virus pneumonia (with aciclovir).
-

PREVENTION:

1. Influenza Vaccination

- Influenza vaccines have been used since the 1940s.
- They are effective in preventing influenza A and B virus infections.
- Inactivated influenza vaccine is effective even in young children, including those under 2 years.
- During the 2009 H1N1 pandemic, a monovalent vaccine was developed and widely used.
- Neuraminidase inhibitors (e.g., oseltamivir, zanamivir) can also be used prophylactically to prevent influenza infection.

2. Respiratory Syncytial Virus (RSV)

- Severe RSV infections in high-risk neonates can be prevented with palivizumab, a humanized monoclonal antibody.
- It is given during RSV epidemics.
- Palivizumab reduces RSV-related hospital admissions by about 50% in premature infants.
- Despite many attempts, an effective RSV vaccine has not yet been successfully developed, though live-attenuated vaccines are under study.

3. Adenovirus

- An oral vaccine against adenovirus types 4 and 7 has been used successfully in military trainees, with about 95% efficacy.
- Production of this vaccine was stopped in 1996 due to manufacturing issues.

4. Pneumococcal Conjugate Vaccine

- Pneumococcal conjugate vaccine has been shown to prevent about one-third of viral pneumonia cases.
 - This effect is likely due to prevention of secondary bacterial coinfection.
-

NUTRITIONAL MANAGEMENT:

1. Impact of Pneumonia on Nutritional Status

The pneumonia significantly increases metabolic stress. Fever, inflammation, and increased work of breathing raise energy expenditure. At the same time

- Appetite is reduced
- Oral intake often declines
- Catabolism increases
- Risk of malnutrition rises
- Malnutrition is associated with:
- Impaired immune response

2. Energy Requirements

- Increased basal metabolic rate
- Systemic inflammatory response
- However, it cautions against overfeeding, particularly in severe cases, as excess caloric intake may increase carbon dioxide production and worsen respiratory burden.

3. Increase Protein Requirement

- Protein intake should be about 1.2–1.5 g/kg body weight per day to support healing and maintain muscle mass.
 - Include High-Quality Protein Sources
 - Provide foods rich in good quality protein such as:
 - Eggs
 - Milk and milk products (milk, curd, paneer)
 - Lean chicken or fish
 - Pulses and dals
 - Soy products (soybean, tofu)
 - Sprouted legumes
 - Use Soft and Easily Digestible Protein Foods
-

INITIAL ASSESSMENT OF THE PATIENT:

PATIENT INFORMATION:

Chief Complaint:

- Cough
- Difficulty in breathing
- Chest discomfort
- Weakness

Duration of pain:

- Chest discomfort from past 7 days

Nature of pain:

- Dull aching pain in the lower chest region
- Pain increases during coughing
- Non-radiating pain
- Associated with productive cough

History on present illness:

The patient is a known case of recurrent lower lobe pneumonia for the past 8 years. He presented to the hospital with complaints of persistent cough, breathlessness, and chest discomfort for the past one week. The cough is productive and associated with fatigue. The patient is also a known case of diabetes mellitus and hypertension and is on regular medication for the same. Due to worsening of symptoms, he was admitted to the hospital for further management and treatment.

Medical consultation and Investigation:

- The patient was evaluated by the physician and advised the following investigations:
 - Complete Blood Count (CBC)
 - Blood sugar levels (RBS/FBS)
 - Chest X-ray (showing lower lobe involvement)
 - Sputum examination
 - Renal function tests
 - Blood pressure monitoring
-

Past medical History:

- Diabetes
- Hypertension

Family history: Nothing significant

Personal History:

1 Diet: Normal

2 Appetite: Normal

3 Smoking: Nil

4 Allergies: Nil

5 Alcohol: Nil

6 Bowel/Bladder: Normal

NUTRITIONAL SCREENING:

- **NAME:** Mr XYZ
- **AGE:** 65 yrs
- **GENDER:** MALE
- **DATE OF ADMISSION:** 21-02-2026
- **TIME OF ADMISSION:** 4:39 pm
- **DURATION OF STAY:** 3 DAYS

ANTHROPOMETRY:

- **HEIGHT:** 168 cm
- **WEIGHT:** 77 kg
- **BMI:** 27.7 kg / m²
- **IBW:** 64 kg
- **ACTIVITY:** SEDENTARY

FOOD HABITS:

- NON VEGETARIAN
-

MEDICAL DIAGNOSIS:

- Lower lobe pneumonia + diabetes mellitus – underwent septoplasty

VITALS:

COMPONENTS	REPORTED VALUE	REFERENCE VALUE
BLOOD PRESSURE	140/80 mmHg	90-120 / 60-80 mmHg
BODY TEMPERATURE	97f – 99 f	97f – 99 f
RESPIRATORY RATE	18 -20 b/m	12-20 b/m
PULSE	100 bpm	60-100 bpm

BIOCHEMICAL PARAMETERS:

PARAMETERS	VALUE	NORMAL RANGE
HB	13.5g/dl	13.0 - 17.0g/dl
RBC	4.55 m/ μ	4.5 - 5.5 m/ μ
WBC	11.30 thou/	4.0 - 10.0 thou/
PLATELET	194 thou/	150 - 410 thou/
HEMATOCRIT	40.4%	40 - 50%

DRUG THERAPY:

SL NO.	Drug Used	Functions
1	Inje- Meropenem {1gm IV}	Used for severe bacterial infection
2	Tab-For can 150 mg	Treats fungal infection
3	Tab- Beco rules	Helps nerve health and recovery
4	Cipro Nasa wash	Clears sinus infection
5	Insulin 4u,5u,14u-	Given because RBS was high

DIABETIC FLOW CHART:

DATE	TIME	BLOODSUGAR LEVEL mg/dl	INSULIN	DOSAGE
21-02-2026	7:10 pm	113	-	-
22-02-2026	6:00 am	160	-	-
22-02-2026	6:00 am	247	-	-
22-02-2026	7:00 am	200	-	-
23-02-2026	6:00 am	172	Act rapid	4 Units
23-02-2026	1:00 am	298	Act rapid	10 Units
23-02-2026	6:00 am	124	-	-
24-02-2026	6:00 am	188	Act rapid	4 Units
24-02-2026	1:00 am	214	Act rapid	5 Units
24-02-2026	6:00 pm	319	Act rapid	14 Units
25-02-2026	6:00 am	149	-	-

**DIAGNOSIS: LOWER LOBE PNEUMONIA + DIABETES MELLITUS –
UNDERWENT SEPTOPLASTY**

24 HOUR DIETARY CALL:

MEAL	MENU
EARLY MORNING	TEA
BREAKFAST	IDLI SAMBAR
MID MORNING	TEA
LUNCH	RICE + DAL + VEG SABZI
MIDAFTERNOON	TEA
DINNER	CHAPATI+VEG SABZI

DETAILED MENU PLAN

Meal	Menu	Ingredients	Amount	Energy [kcal]	Protein [g]	Fat [g]	CHO [g]	Fiber [mg]
Early morning	Tea	Milk	100	73	3.26	4.48	4.94	0
Breakfast	Idli Sambar	Rice	30	105	2.34	0.17	23.15	1.12
		Black gram dal	30	97	6.92	0.51	15.3	3.58
		Toor dal	30	99	6.47	1.59	14.02	4.55
		Onion	10	5	0.15	0.02	0.96	0.25
		Tomato	10	2	0.09	0.05	0.27	0.17
Mid-morning	Tea	Milk	100	73	3.26	4.48	4.94	0
Lunch	Rice	Rice	50	176	3.91	0.28	38.58	1.87
	Dal	Moong dal	30	97	5.2	0.40	13.0	3.50
	Sabzi	Carrot	15	4	0.14	0.83	1.11	0.07
		Capsicum	10 g	4.80	0.14	0.19	0.16	0.09
		Onion	35	0.30	3.1	4	3	0.25
Evening	Tea	milk	100	73	3.26	4.48	4.94	0
Dinner	Chapati	Wheat Flour	60	192	6.34	0.92	38.5	0.92
		Carrot	30	99	6.47	1.59	14.02	4.55
		capsicum	30	99	6	1	14	4
		Onion	10	5	0.15	0.02	0.96	0.25
		Tomato	10	2	0.09	0.05	0.27	0.17
			Total	1220	46.22	38.02	168.61	5.89

3 DAYS HOSPITAL DIET RECALL

DAY 1 - DIABETIC LESS SALT NORMAL DIET

MEAL	MENU
EARLY MORNING	Coffee
BREAKFAST	Idli + Chutney+ Sambar
MID MORNING	Tea
LUNCH	Chapati + sabzi + gravy
MIDAFTERNOON	Tea
DINNER	Chapati + sabzi + gravy

TIMING	MEAL	MENU	QUANTITY
6:30 am-7:00 am	EARLY MORNING	S/L TEA	150 ml
8:00 am-9:00 am	BREAKFAST	IDLI + CHUTNEY+ SAMBAR	3 idly + ¼ cup chutney + ¾ cup sambar
10:00 am-10:30 am	MIDMORNING	S/L TEA	150 ml
12:00 pm-01:00pm	LUNCH	CHAPATI + SABZI + GRAVY	2 chapati ½ cup sabzi ¼ gravy
3:30 pm-4:30 pm	AFTERNOON	S/L COFFEE	150 ml
6:00pm-6:30pm	EVENING	VEGETABLE SOUP	1 small bowl
7:30pm-8:30pm	DINNER	CHAPATI +SABZI + GRAVY	2 chapati + ½ sabzi ½ cup gravy
9:00pm-9:30pm	BEDTIME	S/L MILK	200 ML

DAY 2 - DIABETIC LESS SALT NORMAL DIET

MEAL	MENU
EARLY MORNING	Coffee
BREAKFAST	Vegetable upma +Chutney+ Sambar
MID MORNING	Tea
LUNCH	Chapati + sabzi + Rassam
MIDAFTERNOON	Tea
DINNER	Chapati + sabzi+ gravy

TIMING	MEAL	MENU	QUANTITY
6:30 am-7:00 am	EARLY MORNING	S/L TEA	150 ml
8:00 am-9:00 am	BREAKFAST	UPMA + CHUTNEY+ SAMBAR	1 full bowl ¼ cup chutney + ¾ cup sambar
10:00 am-10:30 am	MIDMORNING	S/L TEA	150 ml
12:00 pm-01:00pm	LUNCH	CHAPATI + PALYA + GRAVY	2 chapati ½ cup sabzi ¾ rassam
3:30 pm-4:30 pm	AFTERNOON	S/L COFFEE	150 ml
6:00pm-6:30pm	EVENING	VEGETABLE SOUP	1 small bowl
7:30pm-8:30pm	DINNER	CHAPATI + PALYA + GRAVY	2 chapati + ½ palya ½ cup rassam
9:00pm-9:30pm	BEDTIME	S/L MILK	200 ML

DAY 3 - DIABETIC LESS SALT NORMAL DIET

MEAL	MENU
EARLY MORNING	Coffee
BREAKFAST	Wheat dosa+ Chutney+ Sambar
MID MORNING	Tea
LUNCH	Chapati + sabzi + Dal
MID AFTERNOON	Tea
DINNER	Chapati +bhindi + gravy

TIMING	MEAL	MENU	QUANTITY
6:30 am-7:00 am	EARLY MORNING	S/L TEA	150 ml
8:00 am-9:00 am	BREAKFAST	DOSA+ CHUTNEY+ SAMBAR	3 dosa ¼ cup chutney + ¾ cup sambar
10:00 am-10:30 am	MIDMORNING	S/L TEA	150 ml
12:00 pm-01:00pm	LUNCH	CHAPATI + SABZI + DAL	2 chapati ½ cup sabzi ½ dal
3:30 pm-4:30 pm	AFTERNOON	S/L COFFEE	150 ml
6:00pm-6:30pm	EVENING	VEGETABLE SOUP	200 ml
7:30pm-8:30pm	DINNER	CHAPATI + BHENDI + GRAVY	2 chapati + ½ bhindi ½ cup gravy
9:00pm-9:30pm	BEDTIME	S/L MILK	200 ML

CALCULATIONS:

HT- 168 {5f, 6inch}

WT -77

BMI: 27.2

WT IN KG- 77

HT IN m2 – 1.67

➤ 1feet – 12inch

➤ 1inch – 2.54

$$5 \times 12 = 60 + 6 = 66$$

$$66 \times 2.54 = 167 \div 100 = \mathbf{1.67}$$

$$1.67 \times 1.67 = \mathbf{2.78}$$

$$77 \div 2.78 = \mathbf{27.69}$$
 Patient is over weight

STEP-1

IBW =

➤ **BROCAS-** Ht in cm – 100

$$168 - 100 = 68$$

➤ **HAMVIS** – [106 LB FOR FIRST 5 FEET + 6 LB FOR EVERY ADTNL INCH]

$$= 106 + 6 \times 6$$

$$= 142 \text{ lb} \div 2.2$$

$$= 64.54$$

➤ **ENERGY**

EAR VALUE

EAR WT x EAR [kcal/kg/d]

$$65 \times 32$$

$$= 2080$$

➤ **IBW VALUE**

$$\mathbf{64.5 \times 32}$$

$$\mathbf{=2064}$$

➤ **PATIENTS VALUE**

$$77 \times 32$$

$$= 2464$$

According to the calculated energy requirement the kcals are about 2000. but the patients current body weight is 77 and is overweight. However, considering the patients age and overweight condition, this energy intake may be excessive. Therefore, the diet plan is restricted to approximately 1600 kcals/d to help in weight control and better management of diabetes.

➤ **PROTEIN**

EAR

$$0.66 \times 65$$

$$= 42.9 \times 4 = 171.6 \text{ kcal}$$

$$\div 2080$$

$$= 8.25$$

- **$\% 18 \div 100 \times 1600 = 288 \div 4 = 72 \div 77 = 0.9$**

- $\% 20 \div 100 \times 1600 = 320 \div 4 = 80 \div 77 = 1.03$

- $\% 25 \div 100 \times 1600 = 400 \div 4 = 100 \div 77 = 1.29$

➤ **FAT**

- **$18\% \div 100 \times 1600 = 288 \div 9 = 32 \div 77 = 0.41$**

- $\% 20 \div 100 \times 1600 = 320 \div 9 = 35.5 \div 77 = 0.46$

➤ **CARBOHYDRATES**

$$C + P + F = 100$$

$$C + 18 + 18 = 100$$

$$C = 36 - 100 = 64$$

$$64 \div 100 \times 1600 = 1024 \div 4 = 256 \text{ g}$$

THUMB RULE:

C: P: F

64:1

NUTRITIONAL GUIDE LINES:

1. Energy

- Provide moderate calories (about 25–30 kcal/kg body weight/day) to meet energy needs and prevent weakness.
- Soft and easily digestible foods are preferred because pneumonia patients may feel tired or have poor appetite.

2. High Protein Diet

- Protein helps repair lung tissue and maintain muscle.
- Give 1.0–1.2 g protein/kg body weight/day.
- Dal and pulses
- Paneer or low-fat milk
- Chicken or fish (if non-vegetarian)
- Sprouts

3. Carbohydrates (for diabetes)

- Carbohydrates should be controlled and evenly distributed through the day to prevent blood sugar spikes.

Preferred foods:

- Whole wheat roti
- Brown rice (small portion)
- Oats
- Millets (ragi, jowar)
- Refined flour food

4. Vitamins and Minerals

- Important for immunity and healing.

Vitamin C foods:

- Orange
- Guava

Vitamin A foods:

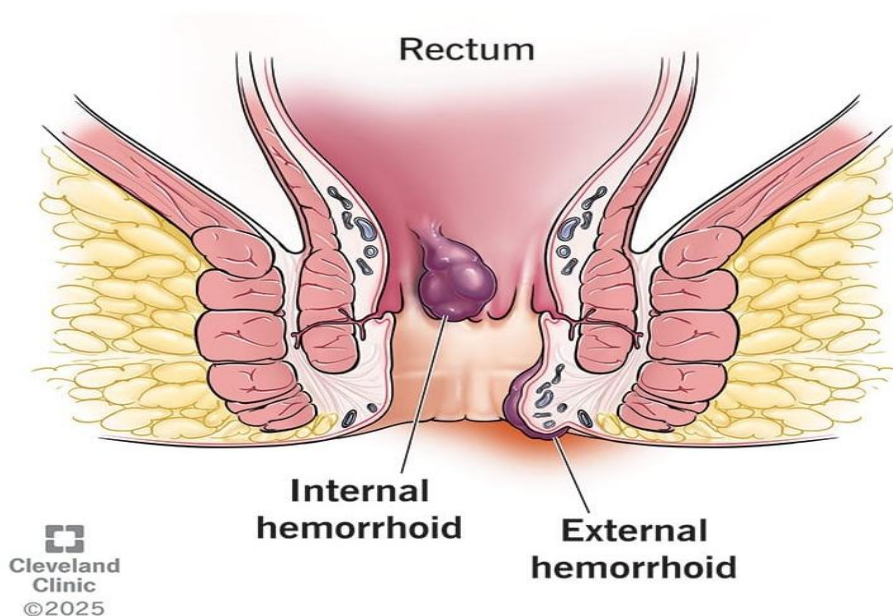
- Carrot
- Spinach
- Pumpkin

5. Limit Fat and Fried Foods

- Heavy oily foods may cause indigestion and discomfort.
Prefer:
 - Steamed
 - Boiled
 - Lightly cooked
-

CASE STUDY – 2

HAEMORRHOIDS



INTRODUCTION:

Haemorrhoids are pathological changes of the normal anal cushions, which contributes to stool evacuation and maintenance of anal continence. The condition develops when there is disruption of the supporting connective tissues within the anal cushions, resulting in enlargement and distal displacement of vascular plexus. The aetiology of haemorrhoids is multifactorial and is commonly associated with factors that increase intra-abdominal and anal canal pressure, such as chronic constipation, prolonged straining during defecation, low fiber diet, sedentary life style and aging. The enlargement or sliding of haemorrhoidal tissue leads to clinical manifestations including rectal bleeding, swelling, prolapse, mucus seepage, and irritation of the perianal skin. In more severe cases, thrombosis of the haemorrhoidal vessels may occur producing significant pain. Due to its high prevalence and impact on quality of life, haemorrhoidal disease remains a common condition encountered in colorectal and general clinical practice. Haemorrhoids are classified as internal or external depending on their location in relation to the dentate line. Internal haemorrhoids are located above the dentate line and are usually pain less but may present with bleeding and prolapse.

ETIOLOGY:

The aetiology of haemorrhoids is multifactorial. Common contributing factors include chronic constipation, prolonged straining during defecation. Haemorrhoids are generally caused by a disturbance of bowel evacuation. This disturbance may occur in the form of constipation or diarrhoea and is closely related to dietary intake. The most common contributing factor is constipation due to lack of fibre and water in the diet. These factors lead to abnormal straining and changes in the anal cushions, which contribute to the development of haemorrhoids.

Clinically, patients commonly present with painless rectal bleeding, swelling, itching, mucus discharge, prolapse during defecation, and irritation of the perianal skin. In severe cases, thrombosis and strangulation may occur, leading to acute pain and complications.

Factors causing Haemorrhoids

- Chronic constipation
- Prolonged straining during defecation
- low fibre diet
- Prolonged sitting (especially on the toilet)
- Obesity
- Aging
- Chronic diarrhoea

PATHOPHYSIOLOGY:

Immediately above the dentate line, the mucosa of the anal canal covers three cushions of stroma that contain arteriovenous fistulae and contribute to continence. Normally, these cushions remain in position; however, disturbance of bowel evacuation, such as constipation or diarrhoea, leads to straining. As these cushions descend, they enlarge and project into the anal canal, forming internal haemorrhoids.

The degree of descent determines the clinical stage:

- Initially causing bleeding (First stage).
 - Prolapsing during defecation (Second stage)
 - Requiring manual reduction (Third stage)
 - Eventually becoming irreducible with disruption of the dentate line (Fourth stage)
-

SYMPTOMS:

- Bleeding that may occasionally lead to anaemia
- Awareness of a lump after defaecation
- Lump requiring manual replacement
- Irreducible lump
- Discharge from the anal region
- Moist perianal area
- Perianal irritation
- Feeling of incomplete bowel evacuation

DIAGNOSIS:

The diagnosis of haemorrhoids is primarily based on a thorough clinical assessment, including patient history, physical examination, and, when necessary, additional investigations.

A detailed history is taken to identify symptoms such as bleeding during defecation, pain, itching, swelling, mucus discharge, or a feeling of incomplete evacuation. Risk factors such as constipation, diarrhoea, prolonged sitting, pregnancy, heavy lifting, and family history are also considered. Physical examination includes visual inspection to detect external haemorrhoids or prolapsed internal haemorrhoids, and a digital rectal examination to feel for internal swelling, tenderness, or abnormalities.

TREATMENT:

Once it has been established that a patient's symptoms are attributed to haemorrhoids, treatment options are discussed. For many patients with intermittent and minimally troublesome symptoms, no active treatment may be required, and reassurance regarding the benign nature of the condition may be sufficient. When the patient seeks relief, management options range from conservative measures to surgical excision, depending on the severity of symptoms and the extent of prolapsing tissue.

Initial management includes advice to avoid straining at stool and to improve bowel function by increasing fluid and fibre intake or by using laxatives. Fibre supplementation has been shown to significantly reduce bleeding. Various topical ointments and containing low-dose anaesthetics, steroids, protectants, antiseptics, and astringents are commonly used, although there is little evidence demonstrating significant long-term benefit.

PREVENTION:

Preventing haemorrhoids mainly involves reducing pressure on the rectal veins and maintaining healthy bowel habits:

- **Avoid straining during bowel movements:** Straining increases pressure in the anal veins, which can lead to haemorrhoids. Respond promptly to the urge to defecate instead of delaying.
- **Irregular bowel habits:** A consistent routine helps prevent constipation and reduces the risk of developing haemorrhoids.
- **Avoid prolonged sitting or standing:** Sitting for long periods, especially on the toilet, increases pressure in the rectal area. Taking breaks and moving regularly can help.
- **Exercise regularly:** Physical activity improves bowel function, reduces constipation, and promotes healthy circulation.

NUTRITIONAL MANAGEMENT:

- Diet plays a crucial role in both preventing and managing haemorrhoids by ensuring soft stools and reducing straining.
 - **High-fibre diet:** Include fruits, vegetables, whole grains, legumes, and nuts. Fiber softens stools, making them easier to pass and reducing pressure on anal veins.
 - **Adequate hydration:** Drink plenty of water throughout the day. Fluids help fiber work effectively and prevent constipation.
 - **Smaller, frequent meals:** Eating in moderation can improve digestion and reduce straining during bowel movements.
 - **Avoid constipating foods:** Limit intake of processed foods, refined flour, excessive dairy, and fried foods that can harden stools.
 - **Smaller, frequent meals:** Eating in moderation can improve digestion and reduce straining during bowel movements.
 - **Probiotics (optional):** Yogurt or fermented foods can support healthy gut flora, aiding digestion and regular bowel movement
 - **Include natural laxatives** – Foods like prunes, figs, and flaxseeds can help prevent constipation.
 - **Limit caffeine and alcohol** – These can dehydrate the body and worsen constipation.
 - **Balanced diet with adequate nutrients** – Vitamins C, E, and zinc help maintain healthy blood vessels and tissue repair.
-

INITIAL ASSESSMENT OF THE PATIENT:

PATIENT PROFILE:

Chief Complain:

- Bleeding per rectum
- Pain during defecation
- Loose stools

Duration of pain: Since last year

Nature of pain: > Pain is localized to the anal region

History of present illness:

The patient has a history of bleeding per rectum for approximately one year. She sought medical advice at a local clinic where she was prescribed medications for about 10–15 days, which provided temporary relief. However, symptoms recurred and gradually worsened, with increased severity of pain and continued bleeding. Due to persistent symptoms and worsening pain, she consulted at Vikram Hospital, where further evaluation was done and surgical management was advised.

The patient also complained of discomfort during defecation and difficulty performing daily activities due to pain. On further assessment, she reported episodes of constipation and straining while passing stools, which aggravated her symptoms. Clinical investigations and physical examination were carried out to identify the underlying cause and assess the severity of the condition. She was advised regarding dietary modifications, adequate fluid intake, and proper bowel habits to prevent further complications

Medical consultation and Investigation:

- Detailed clinical examination was performed.
 - Per rectal examination suggested haemorrhoidal pathology with sphincter involvement.
 - Routine pre-operative investigations were conducted, including:
 - Complete blood count
 - Blood sugar levels
 - Urine routine examination
-

Personal History:

1 Diet: Normal

2 Allergies: Allergic to brinjal and Capsicum

3 Sleep: Adequate

4 Bowel/Bladder: Loose Stools

NUTRITIONAL SCREENING:

- **PATIENT PROFILE:**
- **NAME:** PQR
- **Age:** 26 yrs
- **Gender:** Female
- **Occupation:** House maker
- **Date of Admission:** 16-02-2026
- **Time of Admission:** 16:29 pm
- **Duration of stay:** 2 days

ANTHROPOMETRIC MEASURES:

- **Height:**150 cm
 - **Weight:**49.7kg
 - **BMI:**22kg/m²
 - **IBW:**43- 47
 - **Activity:** Sedentary
 - **Food Habits:** Vegetarian
-

MEDICAL DIAGNOSIS: EXTERNAL HAEMORRHOIDS UNDERWENT LASER HAEMORRHOIDECTOMY

VITALS:

COMPONENTS	REPORTED VALUE	REFERENCE VALUE
BLOOD PRESSURE	100/60 mmHg	90-120 / 60-80 mmHg
BODY TEMPERATURE	97 f	97f – 99 f
RESPIRATORY RATE	20 b/min	12-20 b/m
PULSE	80 / min	60-100 bpm

BIOCHEMICAL PARAMETERS:

PARAMETERS	VALUE	NORMAL RANGE
HB	13.5 g/dl	13.0 - 17.0g/dl
RBC	4.25 m/ μ	4.5 - 5.5 m/ μ
WBC	7.3 thou/ μ	4.0 - 10.0 thou/ μ
PLATELET	369thou/ μ	150 - 410 thou/ μ
HEMATOCRIT	37.1	40 - 50%
RBS	118 mg/dl	, <140 mg/dl

DRUG THERAPY:

SL NO.	Drug Used	Functions
1	INJ Supacef 1.5	Antibiotic used to treat and prevent bacterial infections.
2	INJ Metrogyl 50	Antibiotic effective against anaerobic bacteria and helps control infections
3	INJ Pct	Used to reduce fever and relieve mild to moderate pain.
4	INJ Dynaper	NON- steroidal anti- inflammatory drug used to relieve pain and reduce inflammation.
5	INJ Pan	Reduce stomach acid and helps prevent gastric irritation and ulcers

DIAGNOSIS: HAEMORRHOIDS**24 HOURS DIETARY CALL:**

MEAL	MENU
EARLY MORNING	Coffee
BREAKFAST	DOSA+ SAMBAR
MID MORNING	COFFEE
LUNCH	RICE+ DAL+ SABZI
MIDAFTERNOON	BUTTER MILK
DINNER	CHAPATI VEG SABZI

DETAILED MENU PLAN

Meal	Menu	Ingredients	Amount [g]	Energy [kcal]	Protein [g]	Fat [g]	CHO [g]	Fiber [g]
Early morning	Coffee	Milk	100	73	3.26	4.48	4.94	0
Breakfast	Dosa	Rice	30	105	2.34	0.17	23.15	1.12
		Black gram dal	30	97	6.92	0.51	15.3	3.58
	Sambar	Toor dal	30	99	6.47	1.59	14.02	4.55
		Onion	10	5	0.15	0.02	0.96	0.25
		Tomato	10	2	0.09	0.05	0.27	0.17
Mid-morning	Coffee	Milk	100	73	3.26	4.48	4.94	0
Lunch	Rice	Rice	50	176	3.91	0.28	38.58	1.87
	Dal	Moong dal	30	97	5.2	0.40	13.0	3.50
	Sabzi	Carrot	15	4	1.4	0.83	1.11	0.07
		Onion	35	0.30	1.11	4.2	3.11	0.02
Evening	Butter milk	Curd	100	73	3.26	4.48	4.94	0
Dinner	Chapati + veg sabzi	Wheat Flour	60	192	6.34	0.92	38.5	0.92
		Carrot	30	99	6.47	1.59	14.02	4.55
		Cauliflower	30	99	6	1	14	
		Onion	10	5	0.15	0.02	0.96	0.25
		Tomato	10	2	0.09	0.05	0.27	0.17
			Total	1220	46.22	38.02	168.61	5.89

2 DAYS HOSPITAL DIET RECALL

DAY 1 - NIL PER ORAL [NPO]

DAY 2 - CLEAR LIQUID FOR 1 HOUR THEN CONTINUED WITH SOFT DIET

MEAL	MENU
EARLY MORNING	Tender coconut water
BREAKFAST	Cereals
MID MORNING	Fruit Juice
LUNCH	Khichdi + sabzi + curd
MIDAFTERNOON	Fruit Juice
DINNER	Khichdi + sabzi + curd

TIMING	MEAL	MENU	QUANTITY
6:30 am-7:00 am	EARLY MORNING	TCW	200 ml
8:00 am-9:00 am	BREAKFAST	Idli chutney	2 idli + 1/3 cup chutney
10:00 am-10:30 am	MIDMORNING	Fruit juice	200 ml
12:00 pm-01:00pm	LUNCH	Khichdi + carrot sabzi curd	½ cup sabzi ¼ cup curd
3:30 pm-4:30 pm	AFTERNOON	Fruit juice	200 ml
6:00pm-6:30pm	EVENING	Veg Soup	200 ml
7:30pm-8:30pm	DINNER	Khichdi + carrot sabzi curd	½ cup sabzi ¼ cup curd
9:00pm-9:30pm	BEDTIME	Milk	150 ml

CALCULATIONS:

HT:156cm [5 f,1 inch]

WT:49.9kg

BMI: 20.5kg/m²

WT IN KG:49.9

HT IN m: 1.54

➤ 1feet – 12inch

➤ 1inch – 2.54

$$5 \times 12 = 60 + 1 = 61$$

$$61 \times 2.54 = 154 \div 100 = \mathbf{1.54}$$

$$1.54 \times 1.54 = \mathbf{2.37}$$

$$49.9 \div 2.37 = \mathbf{21.05}$$

STEP-1

IBW =

➤ **BROCAS-** Ht in cm – 100

$$156 - 100 = 56$$

➤ **HAMVIS – [100 LB FOR FIRST 5 FEET + 5 LB FOR EVERY ADDITIONAL INCH]**

$$= 100 + 5 \times 1$$

$$= 105 \text{ lb} \div 2.2$$

$$= 47.7$$

➤ **ENERGY**

EAR VALUE

EAR WT x EAR [kcal/kg/d]

$$55 \times 30$$

$$= 1650$$

➤ **IBW VALUE**

$$47.7 \times 30$$

$$= 1431$$

➤ **PATIENTS VALUE**

$$49.9 \times 30$$

$$= 1497$$

The patient is within the normal weight range; therefore, a very high calorie diet is not required. Generally, the average energy requirement for a normal adult range between 1800–2000 kcal/day. After calculating the patient's energy requirement, the values obtained were between 1432 kcal to 1497 kcal, which is approximately around 1650 kcal/day. Since the patient is neither underweight nor overweight, there is no need to provide a very high or low-calorie diet. However, as the patient has undergone a haemorrhoidectomy, providing slightly higher calories within this range can help in better recovery and healing after surgery. Therefore, an energy intake of around 1650 kcal/day can be recommended for the patient.

➤ **PROTEIN**

EAR

$$0.66 \times 55$$

$$= 36.3 \times 4 = 145.2 \text{ kcal}$$

$$\div$$

$$1650$$

$$= 0.08$$

- **% 18 ÷ 100 x 1650 = 297 ÷ 4 = 72 ÷ 49.9 = 1.4**
- **% 20 ÷ 100 x 1650 = 330 ÷ 4 = 82 ÷ 49.9 = 1.65**
- **% 25 ÷ 100 x 1650 = 412 ÷ 4 = 103 ÷ 49.9 = 2.06**

➤ **FAT**

- **18% ÷ 100 x 1650 = 297 ÷ 9 = 33 ÷ 49.9 = 0.66**
- **%20 ÷ 100 x 1650 = 330 ÷ 9 = 36.6 ÷ 49.9 = 0.73**

➤ **CARBOHYDRATES**

$$C + P + F = 100$$

$$C + 18 + 18 = 100$$

$$C = 36 - 100 = 64$$

$$64 \div 100 \times 1650 = 1056 \div 4 = 264 \text{ g}$$

THUMB RULE:

C: P: F

64:18:18

SAMPLE MEAL PLAN:

MEAL	MENU	INGREDIENTS	AMNT (g)	ENERGY [kcal]	PROTEIN [g]	CHO [g]	FAT [g]
EARLY MRNG	BADAM + MILK	ALMONDS	15	91.4	2.76	0.45	8.77
		MILK	100 ml	72.7	3.2	4.94	8.48
BREAKFAST	Upma	Suji	30 g	122	1.04	23.1	0.16
		Carrot	15 g	4.9	0.14	0.83	0.07
		Onion	10 g	4.80	0.19	1.11	0.09
MIDMORNING	DRY FRUIT LADOO	Dates	15 g	77.17	0.59	11.2	0.05
		Almond	15g	91.4	2.76	0.45	8.77
		Walnuts	15 g	100	2.2	1.52	9.64
		Flax seeds	15 g	88.0	3.5	1.02	10.77
		Jaggery	10 g	35.4	0.18	8.48	0.016
LUNCH	VEGETABLE KHICHDI+ CURD	Moong dal	40 g	119	9.14	19.16	0.24
		Rice	50 g	178	3.9	39.12	0.26
		Carrot	20 g	6.65	0.19	1.11	0.09
		Onion	15 g	8.56	0.27	1.43	0.03
		Peas	25 g	20.33	1.81	2.97	0.03
		curd	50 ml	36.48	1.63	2.27	2.24
MIDAFTRNON	FRUIT JUICE+ SNACKS	Water melon	100 ml	54.7	1.33	0.15	0.11
		Boiled Corn	50 g	36.4	0.80	3.49	0.39
DINNER	CARROT PARATHA + PANEER BHURJI	Wheat flour	60 g	192	6.3	38.8	0.88
		Carrot	30 g	15.6	0.25	2.11	0.30
		Coriander	10 g	3.11	0.35	0.42	0.43

		Paneer	30 g	36.4	2.63	14.67	0.56
		Onion	15 g	8.56	0.27	1.43	0.03
BED TIME	Turmeric Milk	Milk	100 ml	72.7	3.2	4.94	8.48
	TOTAL OIL	OLIVE MUSTARD	5ml 5ml	45 45 = 90			10
	TOTAL SUGAR		5 g	20			
TOTAL				1586 kcal	48.56 [g]	200 [g]	50 [g]

NUTRITION GUIDELINES:

Energy Intake

Provide a normal calorie diet (around 1600–1800 kcal/day) depending on the patient's requirement. Since the patient is within the normal weight range, excessive calorie intake is not required. However, adequate energy should be provided to support post-surgical recovery and healing.

Protein Intake

A moderately high protein diet should be given (about 1.0–1.2 g/kg body weight) to promote tissue repair and wound healing after surgery.

Sources include eggs, milk, curd, paneer, pulses, legumes, soy products, fish, and lean chicken.

Dietary Fibre

A high-fibre diet is recommended to prevent constipation and ensure smooth bowel movement, which helps avoid strain on the surgical site.

Include whole grains, oats, vegetables, fruits, and legumes.

Fluid Intake

Encourage adequate fluid intake (2–3 litres/day) unless restricted. Fluids help soften stools and work with fibre to prevent constipation.

Sources include water, soups, coconut water, and buttermilk.

Vitamins and Minerals

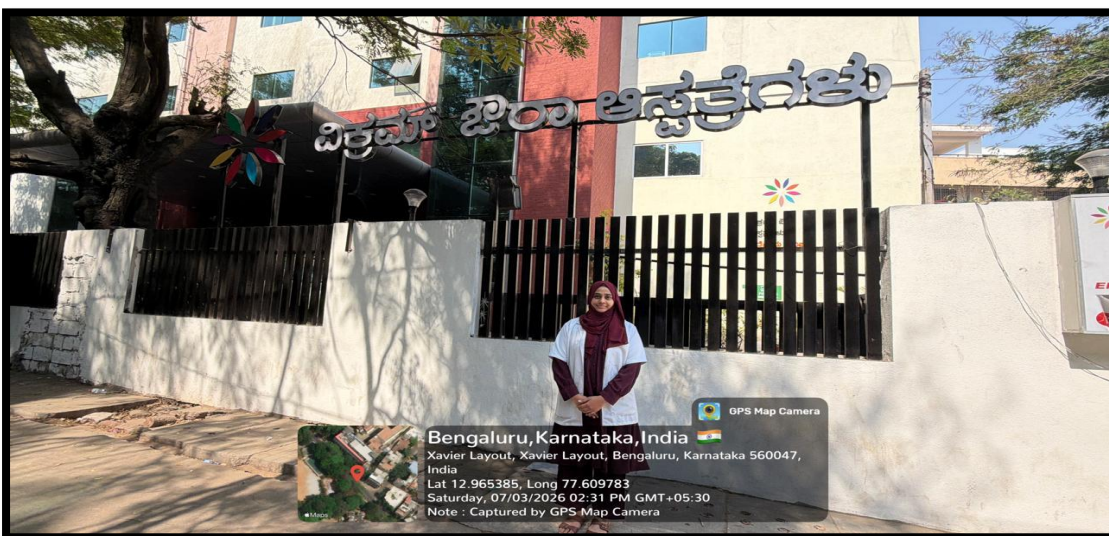
Ensure sufficient intake of vitamin C and zinc as they help in wound healing and immunity.

Sources include citrus fruits, guava, amla, green vegetables, nuts, and seeds.

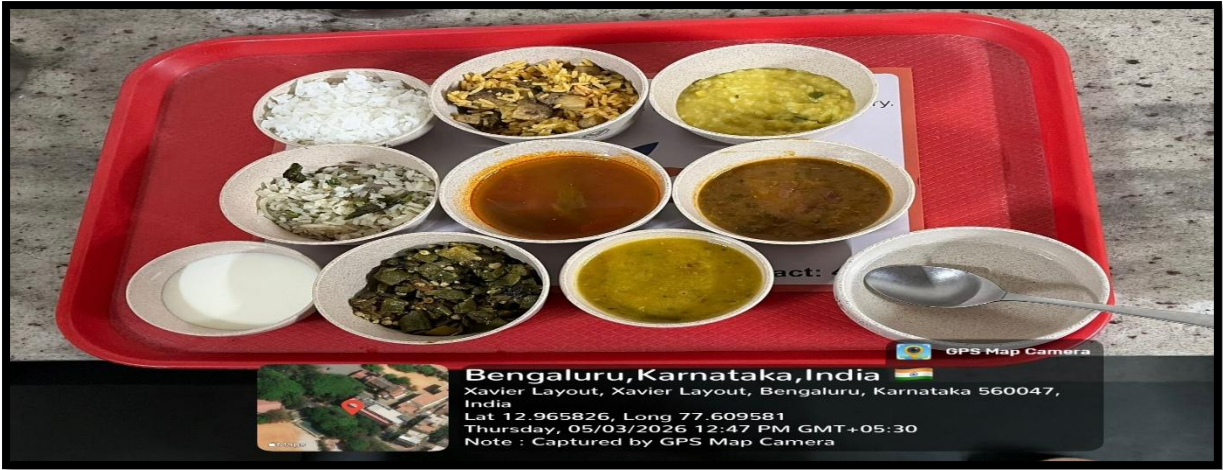
Healthy Fats

Include moderate amounts of healthy fats such as nuts, seeds, and vegetable oils to provide energy and support recovery.

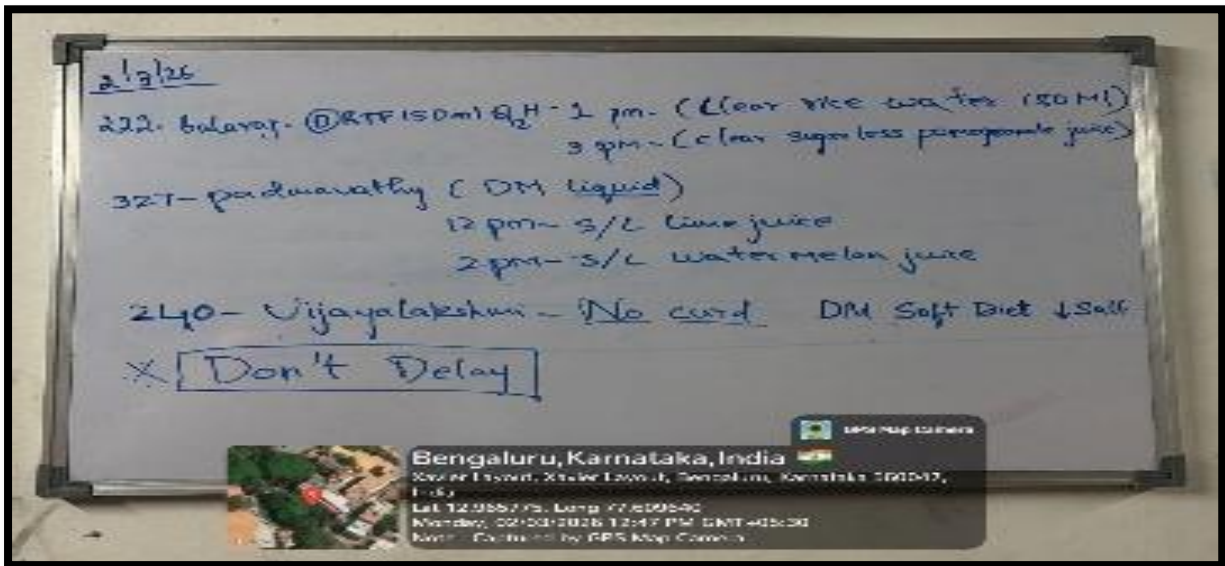
APPENDIX











ANNEXURES

INTERNSHIP LOGBOOK SHEET

STUDENTS NAME: AREENA MARIYAM WAHEED

Name of the Hospital / Institute / Company: VIKRAM AURA [RICHMOND TOWN]

Name of the Mentor / Supervisor / Manager: Dr. AFFRIN NOOR

SL.NO	DATE	TIME IN	TIME OUT	NO OF HOURS	BREIF DESCRIPTION OF THE DAY'S ACTIVITY	Mentor's/ Supervisor's SIGNATURE
01	9/2/26	9:00 am	4:00 pm	7 hr	- Hospital Induction - Understanding Basic terms used in Hospital Dietary Department - And were taken for Hospital Rounds.	Afn
02	10/2/26	9:00 am	4:00 pm	7 hr	- Detailed Explanation of nutritional assessment forms and format. - Observation of the Rpts of the patient & nutritional counselling	Afn
03	11/2/26	9:00 am	4:00 pm	7 hr	- Preparation of nutritional assessment of patients - Assessment of patient and diet card. - Pantry visit and taste testing	Afn
04	12/2/26	9:00 am	5:30 pm	8.5 hr	- Observed diet counselling and diet planning of patients - Food checking and testing, ward rounds	Afn
05	13/2/26	9:00 am	4:00 pm	7 hr	- Observed ward & ICU round and different feeding Practices - Met Medical rep of ensure	Afn
06	14/2/26	9:00 am	5:00 pm	8 hr	- Observed clinical assessment of IP - Biochemical Parameters study for therapeutic diet	Afn
07	16/2/26	9:00 am	4:30 pm	7.5 hr	- Observed case study - Case study allotted - Home work given on Dietary Intervention	Afn

15/2/26
Sunday
Weekly
off

INTERNSHIP LOGBOOK SHEET

SL.NO	DATE	TIME IN	TIME OUT	NO OF HOURS	BREIF DESCRIPTION OF THE DAY'S ACTIVITY	MENTOR'S SIGNATURE
08	17/2/26	9:00 am	4:30 pm	7.5 hr	> Collected details for case study > Went for rounds and nutritional assessment > Check on the Hb level. Food tray checking	Afn
09	18/2/26	9:00 am	5:00 pm	8 hr	> Went for Round ward & ICD > Nutritional assessment > Food tray checking	Afn
10	19/2/26	9:00 am	3:30 pm	6.5 hr	> Went for rounds > Food testing & tray checking > Observed nutritional assessment	Afn
11	20/2/26	9:00 am	3:30 pm	5 hr	> Assisted the chief dietitian for rounds > Documentation taken for case study > Observed nutritional assessment	Afn
12	21/2/26	10:00 am	5:00 pm	7 hr	> Documentation taken for case study > Did Nutritional assessment > Food tray checking > Diet counselling	Afn
13	22/2/26	9:00 am	4:00 pm	7 hr	> Nutrition data entry > Food tray checking > Diet counselling > Nutrition assessment	Afn
14	23/2/26	9:00 am	3:30 pm	6.5 hr	> Diet counselling > Nutrition assessment > Ward rounds > counselled patients for high Protein diet. Under supervision	Afn

INTERNSHIP LOGBOOK SHEET

STUDENTS NAME: AREENA MARIYAM WAHID

Name of the Hospital / Institute / Company: VIKRAM AURA [RICHMOND TOWN]

Name of the Mentor/ Supervisor / Manager: Dr. AFRIN NOOR

SL.NO	DATE	TIME IN	TIME OUT	NO OF HOURS	BREIF DESCRIPTION OF THE DAY'S ACTIVITY	Mentor's/ Supervisor's SIGNATURE
14	24/2/26	-	-	-	Weekly Off	Afn
15	25/2/26	9:00 am	3:30 pm	6.5hr	> Nutrition Assessment > Diet counselling under supervision > Food tray checking	Afn
16	26/2/26	9:00 am	3:30 pm	6.5hr	> Nutrition Assessment > Worked on case study > Food tray checking > Observed diet Planning	Afn
17	27/2/26	9:00 am	3:30 pm	6.5hr	> Nutrition Assessment > worked on case study > Observed diet Planning > Food tray checking	Afn
18	28/2/26	9:00 am	3:30 pm	6.5hr	> Nutrition Assessment > Diet calculation for RTF > Ward rounds > learnt about RTF calculations > learnt about ^{calorie} calorie intake	Afn
19	2/3/26	9:00 am	3:30 pm	6.5hr	> learnt OPD calculation > Water intake calculation > Went for ICU rounds > Individual calorie count	Afn
20	3/3/26	9:00 am	3:30 pm	6.5hr	> Nutritional assessment > Worked on case study > Food tray checking > ICU rounds > Diet planning assessment	Afn

INTERNSHIP LOGBOOK SHEET

SL.NO	DATE	TIME IN	TIME OUT	NO OF HOURS	BREIF DESCRIPTION OF THE DAY'S ACTIVITY	MENTOR'S SIGNATURE
21	4/3/26	9:00 am	3:30 pm	6.5hr	> Went for ward and ICU rounds > Food tray checking > worked on case study > observed diet counselling > Nutrition assessment	Afn
22	5/3/26	9:00 am	3:30 pm	6.5hr	> Went for ICU Round > Food tray checking > Worked on case study > Nutrition assessment	Afn
23	6/3/26	9:00 am	3:30 pm	6.5hr	> Went for ICU Round > Food tray checking > Worked on case study > observed diet counselling > Nutrition assessment	Afn
24	7/3/26	9:00 am	3:30 pm	6.5hr	> Went for ICU Round > Food tray checking > Nutrition assessment > Ward rounds > Diet entry	Afn